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Introduction

The following requirements define the functionality and business rules for an automated audit selection system developed in Python. The system shall identify, shortlist, and prioritize cases for audit through a combination of risk-based methodologies, Survey Interviewer performance indicators, and automated data quality checks. The solution shall support a consistent, transparent, and efficient audit process while ensuring fair coverage across interviewers and minimizing manual intervention.

Project Scope

The automated Risk-Based Audit System (RBA) described in this document shall be implemented for the Household Financial Consumption Survey (HFCS) Wave 2026. The system will support the identification, selection, and prioritisation of cases for audit based on predefined risk criteria, Survey Interviewer performance indicators, and data quality validation checks.

The solution will assist the **Data Collection Unit (DCU)**, **Data Processing Unit (DPU)**, and **Methodology and Quality Unit (MQ)** in strengthening quality assurance processes, improving audit efficiency, and ensuring the reliability and accuracy of survey data collected throughout the HFCS 2026 fieldwork period.

Abbreviations and Acronyms Used in this Document

HFCS – Household Financial Consumption Survey

DCU – Data Collection Unit

DPU – Data Processing Unit

MQ – Methodology and Quality Unit

RBA – Risk-Based Audit System

EAD – Expected Average Duration

Objectives and Context

This document outlines the requirements for an automated case selection and audit support system designed to identify and prioritize cases for audit based on predefined business rules, risk indicators, and data quality measures.

The primary objective of the system is to enhance the efficiency and effectiveness of audit activities by automatically shortlisting cases that present a higher likelihood of errors, inconsistencies, non-compliance, or other data quality concerns. The system will support a risk-based approach to auditing, ensuring that available resources are directed towards cases with the greatest potential impact on data quality and reliability.

The case selection process shall be driven by a combination of:

- **Risk-Based Audits:** Identification of cases based on predefined risk factors, statistical models, thresholds, and historical patterns that indicate a higher probability of data quality issues or respondent misreporting.
- **Survey Interviewer Performance Indicators:** Monitoring and assessment of interviewer performance metrics such as interview duration, response rates, contact attempts, item non-response rates, editing levels, and other operational indicators that may warrant further review.
- **Data Quality Checks:** Automated validation rules and consistency checks designed to detect anomalies, missing information, outliers, logical inconsistencies, unusual response patterns, and other indicators of potential data quality problems.

The system shall provide a transparent and configurable framework that enables audit personnel to define, manage, and update audit criteria as operational requirements evolve. It shall facilitate the generation of prioritized audit lists, support audit decision-making, and contribute to the continuous improvement of survey processes and data quality management.

By integrating risk assessment, interviewer performance monitoring, and data quality validation into a single audit selection framework, the system aims to strengthen quality assurance processes, improve audit coverage, and support evidence-based decision-making throughout the survey lifecycle.

Data Sources

To perform the required audit checks, validations, and report generation activities, the system shall verify and reconcile information from the following data sources:

- **Original Sample File(s)** provided by the Methodology and Quality Unit (MQ).
- **Comprehensive Extract Report**, containing operational and survey data collected during fieldwork.
- **Itemised Telephone Bill**, used to validate contact attempts, call frequency, call dates, and associated telephone numbers.
- **Deaths Central Database**, used to verify whether completed interviews have been conducted with respondents recorded as deceased.
- **Data Checks,** provided by DPU tagging which members require a follow up with regards a data inconsistency

The system shall perform automated cross-checks across these data sources and identify discrepancies, anomalies, or violations of the established business rules. All identified exceptions shall be recorded and presented within the relevant report to facilitate follow-up actions by the DCU, DPU, and MQ units.

Audit Coverage and Case Selection

A minimum of **5% of the net completed cases assigned to each interviewer** shall be selected for audit. The selection process shall be performed in a manner that ensures equitable audit coverage across all interviewers while maintaining the integrity of the risk-based approach.

Commented [EC1]: Still to be confirmed

The Risk Based Audit Procedure shall be cumulative in nature and updated each time the audit selection process is executed. To prevent duplication of work and ensure efficient use of audit resources, the Python process shall automatically exclude:

- Maintain cumulative audit history
- Complete cases that have already been selected in previous audit runs and have a completed survey audit outcome.
- Cases where the audit result status remains blank or incomplete and are still under investigation.

This approach ensures that the same case is not repeatedly selected for audit while an investigation is ongoing or after it has been concluded. It also guarantees a fair and balanced distribution of audit activities across the interviewer pool, allowing new cases to be considered for selection in subsequent runs and maximizing overall audit coverage. The cumulative report shall maintain a complete history of all audited and pending cases to support transparency, monitoring, and reporting requirements.

The shortlisting of cases and Interviewer Performance Review shall be split in two processes, namely, the **Risk based Audit Selection** and the **Interviewer Sample Utilisation Compliance Assessment**

Risk Based Audit Selection

The system will first generate the Total Call Duration linked to the Hh by summing up call durations placed to the sampled numbers associated with that reference from the Itemised Bill document. This duration will then be compared to the Estimated Expected Duration and then assigned a risk score (Refer to Appendix 3) Next, it will shortlist surveys where the actual interview duration is significantly shorter than the estimated duration generated by

the model (Estimated Expected Duration-Refer to Appendix 1). These cases will then be classified according to the severity of the discrepancy between the estimated and actual durations. (Refer to Appendix 2)

The latter shall then be cross matched against the DPU Data Checks Document and Merged.

The duration assessment will be based on a comparison between:

- The survey extract;
- The Expected Interview Duration Model (Refer to Appendix- Illustration 1)
- The itemised telephone bill; and
- Where applicable, the DPU Data Checks output.

DPU Data Check File Format

- Same structure as Survey Extract.
- Contains DPU Flag.
- Contains DPU Check Description.
- Used within Risk-Based Audit Selection process.

The below check shall be performed.

1. Survey Duration from itemised Bill vs Estimated Expected Average Duration

Variables Required

Variable	Source	Purpose
DCU Ref	Survey Extract	Case matching
Interviewer ID	Survey Extract	Interviewer attribution
Interview Date	Survey Extract	Validation
System Recorded Duration	Survey Extract	Duration comparison
Total Call Duration	Telephone Bill	Duration comparison
Expected Duration	Duration Model	Risk assessment

Logic

Condition	Result
Actual duration < 25% of expected duration	Flag as Critical Risk (Code 1)
Actual duration 25%-39% of expected duration	Flag as Very High Risk (Code 2)
Actual duration 40%-59% of expected duration	Flag as High Risk (Code 3)
Actual duration 60%-74% of expected duration	Flag as Medium Risk (Code 4)
Actual duration 75%-89% of expected duration	Flag as Low Risk (Code 5)
Actual duration 90%-120% of expected duration	Flag as Normal (Code 6)
Actual duration > 120% of expected duration	Flag for Complex Case Review (Code 7)

Output

- Household-Level Risk-Based Audit List
- Interviewer Risk Classification Summary

For this exercise, the following will be verified:

Risk-Based Duration Audit Cases: Households identified through the duration-risk methodology.

DPU Data Check Audit Cases: member-level cases requiring follow-up as communicated by DPU, with an indicator showing whether the household was also flagged through the Risk-Based Selection process.

Interviewer-Level Frequency Report: a management report summarising flagged cases by interviewer, including risk category distributions and overlap with DPU Data Checks. This report will support audit planning, resource allocation and the compilation of post-audit findings.

The audit findings shall be presented in four outputs forming Report 1:

1A. Household-Level Risk-Based Audit List

This will generate a list at Household level (same as Household Details Page from Extract). Cases Shortlisted with the Risk based assessment will be listed. To prevent duplicate audit calls, cases appearing in the DPU Data Check Audit List shall be excluded from the Risk-Based Audit List.

1B. Member-Level DPU Data Check Audit List

This will generate a list at Member level (same as Section 1 from Extract). Cases selected by DPU for data checks shall be tagged accordingly, with an additional indicator identifying those also flagged through the Risk-Based Duration Audit methodology.

1C. Long Duration Cases

This report will be at Household Level same as the Risk-Based Audit List, however, this will list the cases Assigned Risk Classification 7- Long Duration Surveys

1D. Interviewer-Level Risk Classification Summary

This shall be generated based on the Risk-Based Duration Audit results. The summary shall present the number of cases classified for each interviewer. The purpose of this summary is to provide a high-level snapshot of interviewers with elevated concentrations of duration-related audit risks and to support the prioritisation of audit activities. Interviewers shall be ranked according to the volume and severity of their flagged cases, enabling a quick identification of the interviewers requiring immediate review.

Interviewer ID	Interviewer	Flagged for data checks	Code 1	Code 2	Code 3	Code 4	Code 5	Code 6	Code 7	Total
0000001M	INT015	3	0	0	1	1	1	3	0	**6**
0000002L	INT008	1	0	0	0	0	1	3	1	**5**
0000004L	INT023	6	1	1	2	1	0	1	0	**6**
0000003M	INT004	2	0	0	2	0	0	1	2	**5**

Where:

Risk Score = (Code 1 × 7) + (Code 2 × 6) + (Code 3 × 5) + (Code 4 × 4) + (Code 5 x3) + (Code 6 x2) + (Code 7 x1)

This provides management with a quick snapshot of interviewers exhibiting the highest concentration of duration-related risk cases and helps prioritise audit and supervisory activities.

The Risk-Based Audit List and the DPU Data Check Audit List shall be sorted in accordance with this ranking, ensuring that cases associated with interviewers presenting the highest concentration of duration-related risks are prioritised for audit. This approach enables auditors to focus their efforts on the highest-risk interviewers first and supports the efficient allocation of audit resources.

Interviewer Sample Utilisation Compliance Assessment

The second process will focus on identifying potential inefficiencies or non-compliance in interviewers' fieldwork practices, specifically assessing whether the sample provided is being utilised appropriately when attempting to contact selected households.

The Compliance assessment will be based on a comparison between:

- The itemised telephone bill; and
- Original Sample from MQ
- The survey extract

The findings shall be presented in four outputs forming Report 2:

2A. Non Contact Cases

The system shall identify all cases assigned a **Non- Contact** disposition codes (31 and 32- refer to Appendix 4) and verify that at least **four call attempts** were made to all sampled telephone numbers associated with the selected household (received from MQ). The call attempts must have been conducted on **different days and times** before the Non- Contact disposition code was assigned in accordance with interviewer instructions and briefing procedures.

Cases that do not satisfy this requirement shall be flagged and included in the report for further review.

Applicability Rule

All sampled contact numbers linked to a household shall be included in the validation. Where a sampled number is no longer present in the Survey Extract, the system shall verify that at least one call attempt was made to that number; otherwise, the number shall be flagged.

A final household-level validation status shall be assigned based on the outcome of all sampled-number checks. If one or more sampled contact numbers are flagged, the household shall be **tagged for review**. Otherwise, the household shall **pass** the validation.

Logic

Condition	**Result**
Any valid sampled number has fewer than 4 attempts	Flag
Sampled number removed from the Survey Extract with an attempt	Exclude from validation

Sampled number removed from the Survey Extract without an attempt	Flag

A Household-Level Contact Compliance Report shall be generated containing all households flagged through this validation process.

The report shall follow the structure of the *Household Details* page within the Survey Extract and shall be sorted in ascending order by **DCU Ref**.

The report shall include all **sampled contact numbers provided by MQ** (landline, CONTACT1_HHLD to CONTACT5_HHLD).

For each sampled contact number, an additional variable shall be included indicating the number of call attempts still required to satisfy the applicable validation rule.

Where a sampled contact number satisfies the validation criteria, the corresponding value shall be recorded as 0.

A household shall be flagged when at least one sampled contact number does not meet the validation requirements. The total number of missing calls is calculated for each household, and the report is sorted alphabetically by Interviewer and then in ascending order by Total Missing Calls.

Example Layout

DCU Ref	Interviewer ID	Interviewer Name & Surname	landline	Missing Calls - landline	CONTACT1_HH	Missing Calls - CONTACT1_HH	CONTACT2_HH	Missing Calls - CONTACT2_HH
60001	000000M	INT001	21234567	0	99123456	1	99887766	0
60002	000000L	INT003	21456789	2	99001122	0	99775544	3

This allows auditors to immediately address both the non-compliant households and the specific contact numbers that did not meet the required call-attempt threshold.

2B. Refusals and Other Reasons

The system shall identify all cases assigned a Refusal or Other Reason disposition codes (42, 43, 44, 53 and 55- refer to Appendix 4) and verify that at least one call attempt was made either to the telephone number recorded in the Phone_Used variable or to at least one sampled contact number associated with the household in the Original Sample provided by MQ.

Cases for which no corresponding call attempt can be identified shall be flagged and included in the relevant audit report for further review.

The report is sorted alphabetically by Interviewer and then by Unattempted Number_1, with cases where Unattempted Number_1 contains a reported number appearing at the top of the list. Cases where Unattempted Number_1 is blank are listed afterwards.

Logic

Condition	Outcome
Call found to Phone_Used	Pass
No call to Phone_Used, but call found to at least one MQ sampled number	Pass

No call found to Phone_Used and no call found to any MQ sampled number	Flag

Sample Report Layout

DCU Ref	Interviewer ID no	Interviewer Name & Surname	Completion_status	Phone_Used	Matching Call Found	Validation Result
60001	0000000M	INT001	Refusal	99123456	No	Flag
60002	0000000L	INT003	Other Reason		No	Flag

2C. Unreachable Cases

The system shall identify all cases assigned Unreachable disposition codes (62 and 63- refer to Appendix 4) and verify that call attempts were made to all sampled telephone numbers associated with the household within the Original Sample provided by MQ.

Any case where one or more sampled telephone numbers (landline, CONTACT1_HHLD to CONTACT5_HHLD, where available) do not have a corresponding call attempt recorded in the Itemised Telephone Bill shall be flagged and included in the report for further investigation.

A Household-Level Unreachable Validation Report shall be generated containing all flagged cases. The report shall follow the structure of the Household Details page within the Survey Extract and shall be sorted in ascending order by DCU Ref.

Logic

Condition	Outcome
All sampled telephone numbers have at least one recorded call attempt	Pass
One or more sampled telephone numbers have no recorded call attempts	Flag for Review

Sample Report Layout

DCU Ref	Interview er ID	Interview er Name and Surname	Completion_status	landline	CONTACT1_HHLD	CONTACT2_HHLD	CONTACT3_HHLD	CONTACT4_HHLD	CONTACT5_HHLD	Unattempted Number_1	Unattempted Number_2	Unattempted Number_3	Unattempted Number_4
60001			Unreachable	2E+07	1E+08	1E+08	-	-	-	None			

60002			Unreachable	2E+07	1E+08	1E+08	1E+08	-	-	CONTACT3_HHLD	CONTACT4_HHLD		

2D. Interviewer Contact Compliance Frequency Table

This report shall be generated summarising the results of the above checks at **Interviewer Level**.

The report shall:

- Aggregate all cases flagged through the Contact Compliance
- Categorise the reason(s) for which each case was flagged.
- Rank interviewers according to the volume of flagged cases.

An example structure is shown below:

Interviewer ID Card	Interviewer Name & Surname	Non-Contacts with Insufficient Attempts	Refusals without Recorded Calls	Unreachables with Unattempted Numbers	Categories falgged	Total Flagged Cases
000000M	INT001	5	2	1	3	8
000000L	INT002	6	0	0	1	6
000000H	INT003	4	1	0	2	5

The purpose of this report is to quickly identify interviewers requiring further review based on Non Response Surveys, guidance, or corrective action, and to support the preparation of management and audit reports relating to interviewer performance and adherence to fieldwork procedures.

The objective of this second process is to identify interviewers who are not using the sample adequately.

Report 3 - Deceased Respondent Verification Report

The system shall identify surveys completed with respondents whose identification details match records contained within the Deaths Central Database.

A comparison shall be performed using the respondent identification details available within the HFCS Survey Extract and the corresponding records held within the Deaths Central Database.

Any confirmed matches shall be flagged and reported for immediate review to assess the validity of the interview and the accuracy of the respondent information recorded.

A Member-Level Deceased Respondent Verification Report shall be generated containing all matched records. The report shall be presented in a member-level format, with one record per flagged respondent, and shall be sorted in ascending order by DCU Ref.

Logic

Condition	Outcome
No match found in the Deaths Central Database	Pass
Match found in the Deaths Central Database	Flag for Review

Sample Report Layout

DCU Ref	Reference_Number	Person_Number	Name	Surname	Identification Number	Date of Death	Match Status
60001	HFCS260001	1	Joseph	Borg	123456M	45762	Flagged
60015	HFCS260015	2	Maria	Vella	654321M	45605	Flagged

Report 4 - Duplicate Identification Number Report

The system shall identify instances where the same identification card number has been recorded across different households within the HFCS Wave 2026 dataset.

A comparison shall be performed across all respondents included in the HFCS Survey Extract to identify identification card numbers associated with more than one household.

Potential duplicate records shall be flagged and reported for investigation to determine whether they are the result of data entry errors, respondent duplication, household linkage issues, or other data quality concerns.

A Member-Level Duplicate Identification Number Report shall be generated containing all identified duplicate records. The report shall be presented in a member-level format, with one record per respondent, and shall be sorted in ascending order by DCU Ref.

Logic

Condition	Outcome
Identification card number appears in only one household	Pass
Identification card number appears in multiple households	Flag for Review

Sample Report Layout

DCU Ref	Member Reference_Number	Name	Surname	Identification Number	Related Household(s)	Validation Result
60001	60000-1-0-1	Joseph	Borg	123456M	HFCS260145	Flagged
60045	60145-1-0-2	Maria	Vella	123456M	HFCS260001	Flagged
60072	60072-1-2-3	Paul	Camilleri	654321M	HFCS260120, HFCS260310	Flagged

The requirements described above constitute the minimum functionality required for the HFCS 2026 Risk-Based Audit System. The following enhancements are proposed for consideration in future iterations of the solution.

Suggested Enhancements for HFCS Risk-Based Audit System

E-001 Section-Level Duration Risk Assessment

Enhance the duration audit by analysing the time spent in individual questionnaire sections instead of only the overall interview duration.

Benefits

- More targeted audit calls
- Identification of sections potentially rushed during the interview
- Better interviewer coaching and training in future waves

E-002 Itemised Bill Duration vs System Recorded Survey Duration

In addition to Estimated Expected Duration vs Itemised Bill Duration, the system would compare the interview duration derived from the Itemised Telephone Bill against the system-recorded survey duration captured within the HFCS Survey Extract.

The purpose of this check is to identify significant discrepancies between the duration recorded by the survey system and the duration supported by the telephony records.

Benefits

This check helps identify situations where the duration recorded by the survey system is not supported by the telephone records.

Examples include:

- Technical issues resulting in incorrect survey duration recording.
- Incorrect interview start/end times.
- Interviews left open unintentionally.
- Potential falsification or poor recording practices.
- Calls that were much shorter than the recorded interview duration.

A case may appear normal based on the Expected Duration Model, but still be suspicious if:

Source	Duration
Survey Extract	12 minutes
Itemised Bill	75 minutes

This discrepancy would warrant further investigation, even if the 75-minute duration falls within the expected range.

E-003 Survey Date vs Last Recorded Call Date

Benefit

This check verifies whether the interview appears to have been conducted on the date recorded in the survey system.

Examples include:

Survey Date	Last Call Date	Result
46068	46068	Consistent
46068	46063	Requires review
46068	No calls found	Requires review

This validation can help identify:

- Cases where the survey completion date may have been entered incorrectly.
- Cases where no supporting call activity exists near the interview date.
- Potential discrepancies between fieldwork records and telephony records.
- Households that may require additional verification during the audit process.

E-004 Dashboard and Visual Analytics

This enhancement would consolidate information generated by the Risk-Based Audit System into a visual reporting environment, enabling users to monitor project performance, audit progress, interviewer risk profiles, and data quality indicators through dynamic charts, tables, filters.

The dashboard shall support analysis at both project level and interviewer level, allowing management to identify emerging issues quickly and take proactive corrective action where necessary.

Benefits

- Real-time monitoring of audit activities and quality indicators.
- Improved visibility of data quality issues throughout the data collection lifecycle.
- Faster identification of operational and interviewer-related risks.

- Enhanced evidence-based decision-making for DCU, DPU and MQ.
- Improved allocation of audit and supervisory resources.
- Better monitoring of interviewer performance across multiple projects.

Strategic Enhancement

In the longer term, the dashboard could be standardised across surveys and integrated with additional interviewer performance indicators to support the development of an Interviewer Performance and Quality Framework.

Such a framework could contribute towards:

- Annual interviewer performance evaluations.
- Project-specific interviewer assessments.
- Identification of training and coaching needs.
- Recognition and reward mechanisms based on objective quality and performance measures.
- Development of a structured interviewer grading and reward system supported by measurable operational and quality indicators.

This would transform the dashboard from a reporting tool into a broader interviewer quality-management and performance-monitoring platform.

E-005 Audit Outcome Feedback

Use completed audit results to continuously improve future audit selection.

Examples

- Identify patterns common to confirmed problematic cases.
- Increase weighting of indicators linked to confirmed audit findings.
- Refine duration-risk thresholds over time.

Benefits

- More effective risk models
- Better audit efficiency
- Continuous improvement of the audit methodology

E-006 Telephone Attempt Timeline

Generate a chronology of all call attempts for flagged cases. Useful during interviewer investigations.

Conclusion

The HFCS 2026 Risk-Based Audit System (RBA) will provide an automated, transparent, and risk-based framework for selecting and prioritising cases for audit. By combining duration-risk analysis, contact compliance assessments, DPU data checks, and data quality validations, the system will improve audit efficiency, consistency, and coverage while reducing manual effort.

The solution will support MQ, DCU, and DPU in identifying potential quality concerns, monitoring interviewer performance, and directing audit resources towards the areas of greatest risk. In doing so, it will contribute to improved data quality, stronger quality assurance processes, and more informed decision-making throughout the HFCS 2026 fieldwork period.

The proposed framework also provides a foundation for future enhancements, allowing the system to evolve into a broader interviewer performance and survey quality management tool.

Appendices

Appendix 1

HFCS 2026 Expected Survey Duration Model

Section	Description	Eligibility Condition	Loop Condition	Standard Time (Minutes)	Calculation Rule
Cover & Introduction	Introduction, identification and consent	Always	No	2.0	Fixed
Household Components	Household composition and member listing	Always	Number of household members	1.0 per member + 2.0 base	2 + (HH Members × 1)
Section 1 Personal Characteristics	Demographic, education, citizenship, marital status	Member aged 16+	Number of members aged 16+	2.0 per eligible member	Members16Plus × 2
Section 2 Main Residence	Characteristics and ownership of main residence	Always	No	6.0	Fixed
Section 2 Main Residence Loans	Mortgage loans secured on main residence	HB1010 = Yes	Number of loans	4.5 per loan	HB1010 × 4.5
Section 2 Other Properties	Additional real estate owned	HB2410 = Yes	Number of properties	5.0 per property	HB2410 × 5
Section 2 Other Property Loans	Loans secured by additional properties	HB300x = Yes	Number of loans	4.0 per loan	Total Property Loans × 4
Section 3 Credit Constraints	Applications and financial constraints	Applicable responses recorded	Number of applications (HC0340)	2.5 per application	HC0340 × 2.5
Section 3 Loan Rejections	Credit applications rejected	HC0410 = Yes	Number of rejections	2.5 per rejection	HC0410 × 2.5
Section 4 Businesses	Ownership and valuation of businesses	HD0210 = Yes	Number of businesses	7.0 per business	HD0210 × 7

Section 4 Financial Assets	Deposits, securities, investment products	Always	No	7.0	Fixed
Section 5 Employment	Labour market status and employment details	Member aged 16+	Number of eligible members	4.0 per eligible member	Members16Plus × 4
Section 6 Pensions	Pension ownership and pension plans	Pension exists	Number of plans	3.0 per plan	PensionPlans × 3
Section 7 Income	Employment, pension and other income	Income source exists	Number of income recipients	2.5 per recipient	IncomeRecipients × 2.5
Section 8 Inheritances & Gifts	Wealth transfers received	Inheritance/gift exists	Number of inheritances	2.0 per inheritance	Inheritances × 2
Section 9 Consumption	Household expenditure indicators	Always	No	4.0	Fixed
Section 10 Expectations	Expectations, attitudes and perceptions	Always	No	3.0	Fixed
Section 11 Interview Outcome	Interview closure and administrative details	Always	No	2.0	Fixed

Appendix 2

Audit Thresholds

Actual Duration as % of Expected Duration	Risk Classification Code	Risk Classification Label
<25%	1	Critical Risk
25%-39%	2	Very High Risk
40%-59%	3	High Risk
60%-74%	4	Medium Risk
75%-89%	5	Low Risk
90%-120%	6	Normal
>120%	7	Complex Case Review

Commented [EC2]: Interviewers flagged under this category will be subject to further review, particularly where they also appear within the first, second, or third categories.

Commented [EC3]: These cases will also be analysed from a training and performance perspective to identify opportunities for improvement, especially where interviewers may not be using the available sample efficiently and, as a result, are placing an unnecessary burden on households.

Appendix 3

Risk Weighting Table

Risk Code	Risk Classification	Weight	Rationale
Code 1	Critical Risk	7	Represents the highest level of concern and therefore receives the greatest weighting.
Code 2	Very High Risk	6	Indicates a significant discrepancy requiring priority attention.
Code 3	High Risk	5	Indicates a substantial duration-related risk requiring investigation.
Code 4	Medium Risk	4	Represents a moderate risk requiring review.
Code 5	Low Risk	3	Limited concern but contributes to interviewer profiling.
Code 6	Normal	2	Used to provide context regarding interviewer workload and case distribution.
Code 7	Complex Case Review	1	Cases exceeding expected duration, requiring contextual review but representing the lowest risk weighting.

Appendix 4

Completion_Status of the survey_Standard code list

11 - Complete interview by desired respondent(s), i.e. 'reference person' and adult household members
12 - Complete interview: partly by desired respondent(s) and partly by proxy or all by proxy
20 - Partial interview by desired respondents or by proxies
31 - No contact with anyone at sampled dwelling
32 - Contact made at sampled dwelling/ household, but not with any responsible resident known to live at the address
41 - Office refusal
42 - Refusal at introduction/ before interview (either by desired respondent or by proxy)
43 - Refusal during the interview/break-off
44 - Broken appointment, no re-contact
51 - Away/in hospital during survey period
52 - Physically or mentally unavailable/ incompetent/ill at home during the survey period
53 - Language barrier
54 - Cases that were not completed because of end of fieldwork (and not falling under 44 or 61)
55 - Other non-response
61 - Not attempted
62 - Inaccessible
63 - Unable to locate address
64 - Unknown whether address contains residential housing; other unknown eligibility
71 - Not yet built/under construction/ demolished/derelict
72 - Non-residential address/business purpose/communal establishment/ institution
73 - Vacant/empty
74 - Address occupied, but no resident household (not the main residency – it is only used as a secondary home)
75 - Address out of sample / other ineligible
80 - Interview completed but discarded after review

Appendix 5

Variable Reference Catalogue

Business Description	Variable Name	Source
Primary Case Identifier	DCU Ref	Household Details
Address Reference	addr_RefNo	MQ Sample
Survey Outcome Disposition Code	Completion_status	Household Details
Interviewer ID Card	Interviewer_ID_Card	Household Details
Interviewer Name	Interviewer_Name	Household Details
Interviewer Surname	Interviewer_Surname	Household Details
Total Interview Duration	Total_time	Household Details
Sample Landline	landline	MQ Sample and Household Details
Sample Contact Number 1	CONTACT1_HHLD	MQ Sample and Household Details

Sample Contact Number 2	CONTACT2_HHLD	MQ Sample and Household Details
Sample Contact Number 3	CONTACT3_HHLD	MQ Sample and Household Details
Sample Contact Number 4	CONTACT4_HHLD	MQ Sample and Household Details
Phone Used for Interview	Contact5	Household Details
Dialled Numbers	Called Number	Itemised Bill
NSO Extension No	Extension_no	Itemised Bill
Rainbow Call Duration	Duration	Itemised Bill

Commented [EC4]: Might have been renamed, but it is the fifth field referring to contact numbers

Appendix 6

Key Section Applicability Rules

Section	**Main Trigger**
Section 1	Every household member
Section 2	Household exists and respondent ≥16
Section 2 mortgage module	Q2.12 = Yes
Section 2 other properties module	Q2.28 = Yes
Section 3	Respondent ≥16
Section 4	Respondent ≥16
Section 5	Member age ≥16
Section 6	Member age ≥16
Section 7A	Member has employment, self-employment or pension income
Section 7B	Household respondent
Section 8	Household received inheritance/substantial gift
Section 9	Household respondent
Section 10	Household respondent
Section 11	Household respondent + interviewer